# LAB 10

# Ordinary Differential Equation

## Assignment:

**Assignment** –

Read about Modified Euler method and Attempt a numerical based on it.

One numerical as suggested by Faculty.

Sample numerical.

Solve the following initial value problem over the interval from $t = 0$ to where $y(0) = 1$. Display all your results on the same graph.

$$\frac{dy}{dt} = yt^2 - 1.1y$$

**(a)** Analytically.
**(b)** Using Euler's method with $h = 0.5$ and $0.25$.

## 1) Analytical solution

$$\frac{1}{y}dy = (t^2 - 1.1)dt$$

Integrating:

$$\ln\ y = \frac{t^3}{3} - 1.1t + C$$

Using $y(0) = 1$:

$$C = 0$$

So exact solution is:

y=e^(t^3/3−1.1t)

**Modified Euler's Method:**

Instead of approximating f(x, y) by $f(x_0, y_0)$ as in Euler's method. In the modified Euler's method we have the iteration formula

$$y_1^{(n+1)} = y_0 + \frac{h}{2}[f(x_0, y_0) + f(x_1, y_1^{(n)})] \ ,n = 0,1,2 \ldots\ldots$$

Where $y_1^{(n)}$ is the nth approximation to y1 .The iteration started with the Euler's formula

$$y_1^{(0)} = y_0 + hf(x_0, y_0)$$

**Matlab Code**

/MATLAB Drive/untitled13.m

```
% y' = 4e^(0.8t) - 0.5y, y(0) = 2, h = 1, t = 0 to 4
h = 1;
t = 0;
y = 2;
fprintf(' t \t y_euler\n');
fprintf(' %d \t %.5f\n', t, y);
t_vals = t;
y_vals = y;
for t = 0:h:3
    y = y + (4*exp(0.8*t) - 0.5*y) * h;
    t_vals = [t_vals, t+h];
    y_vals = [y_vals, y];
    fprintf(' %d \t %.5f\n', t+h, y);
end

plot(t_vals, y_vals, 'r--s', 'LineWidth', 2);
xlabel('t');
ylabel('y');
title("Euler's Method");
```

*Figure 1 Modified Euler Method*

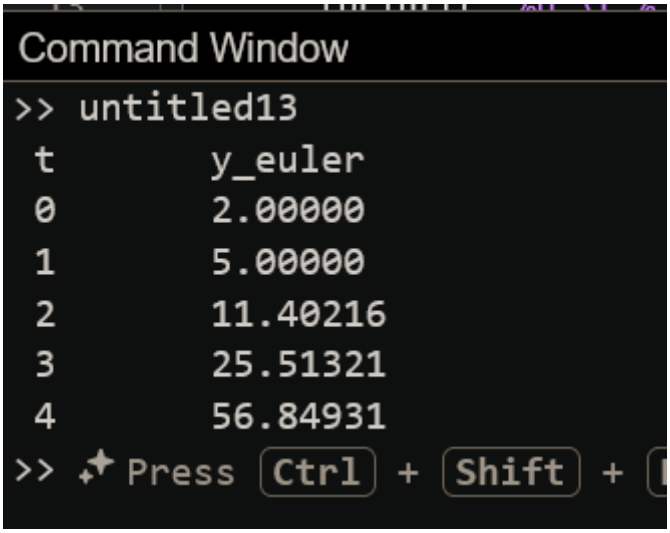


*Figure 2 Result*

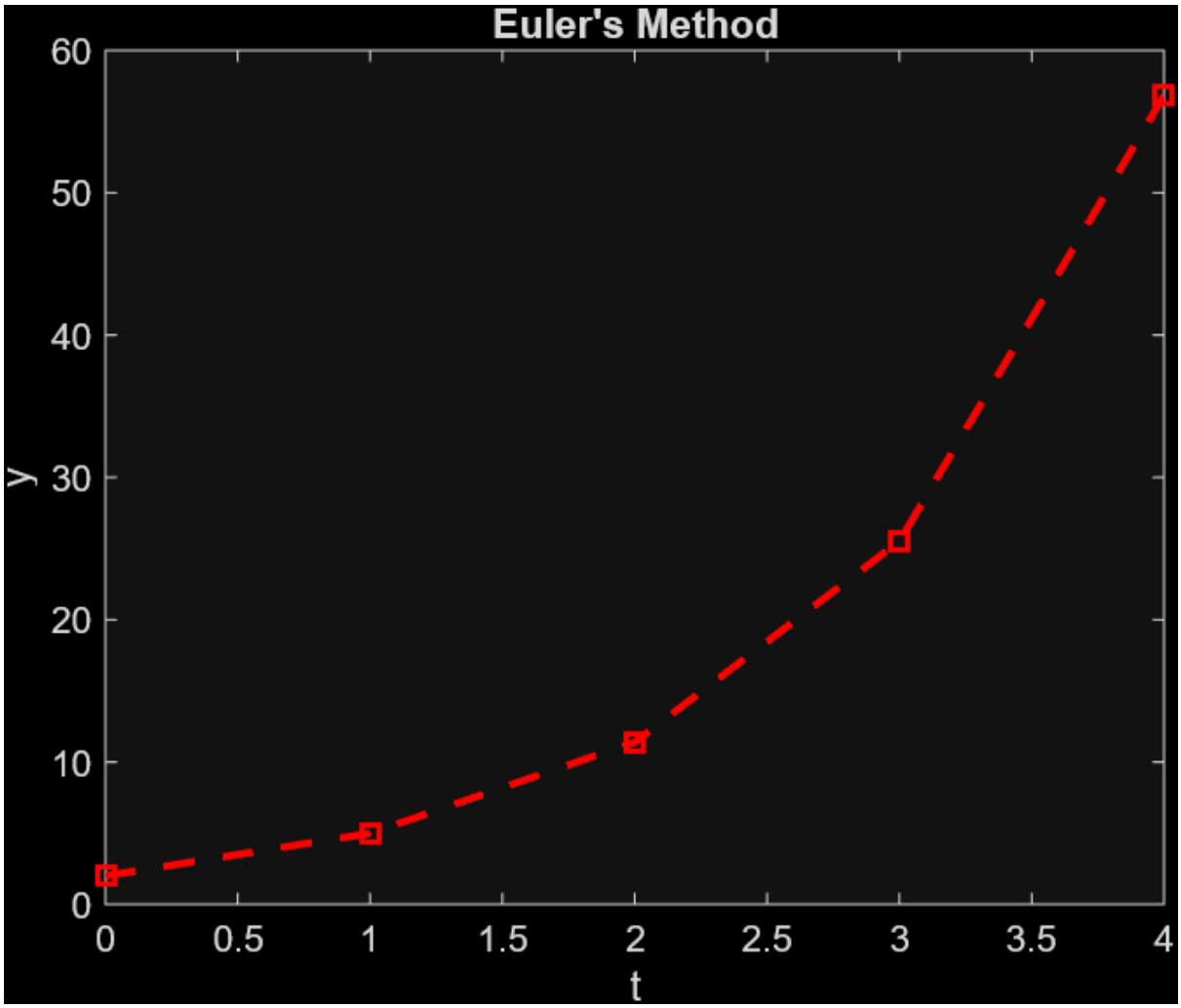


*Figure 3 Euler's method graph of y wrt t*